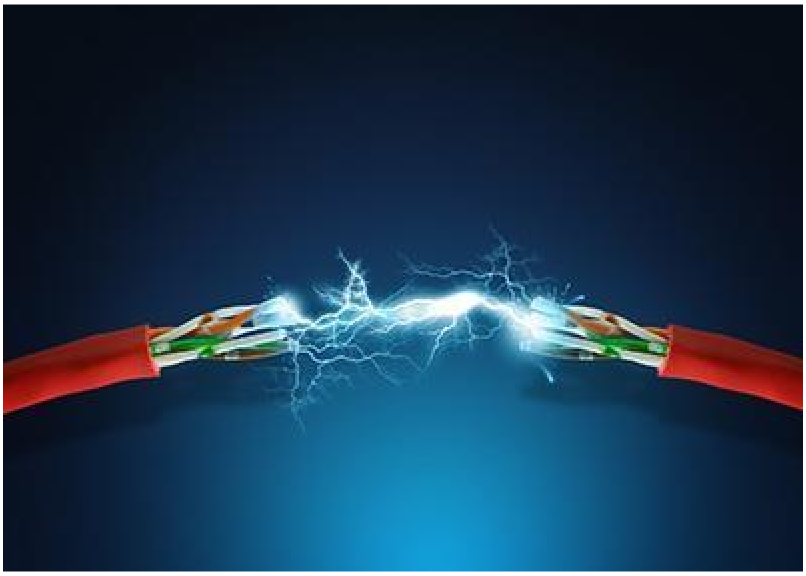
Integrated AFCI Function in Inverter

ENGLISH

V1.0 2024/12/26

This article describes a common electrical feature in photovoltaic systems – arcing, and provides our solution to the hazards posed by arcing.

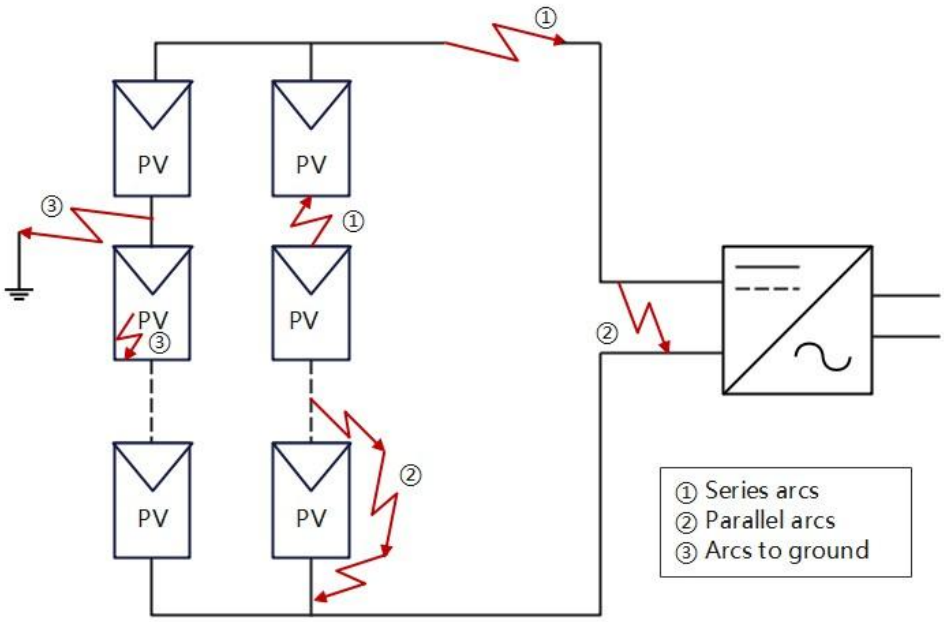
Millions of households and industries around the world have already chosen PV systems to power their loads. PV systems provide a green, economical, stable and convenient source of energy for people's lives, and also try to provide a safe operation environment for users. Among the PV safety incidents that occur worldwide, electrical fires have the highest incidence and cause the most damage. Most of these fire incidents in PV plants are caused by DC arcs, so the necessary protective measures need to be taken to improve the safety of PV systems, and this article introduces one of these measures, the Arc-detection function (AFCI).



1. What's an arc

An electric arc, also known as an arc discharge, occurs when one conductor separates from another in an energised high voltage circuit and an electric arc may appear at both ends. Arcs generate high temperatures and sometimes lead to open flames. In electrical systems, arcing not only causes the surrounding insulation to decompose and become ineffective, but also tends to ignite nearby materials.

DC arcs can be divided into three types, series arcs, parallel arcs and arcs to ground. Due to the electrical structure and the characteristics of PV power generation, among the fires caused by the three types of arcs, the fire caused by the series arc has the highest probability of occurrence, about 80%.



In photovoltaic systems, arcing can be caused by a variety of possibilities, mainly by faulty terminals on the DC side such as loose or separated connections, or by aging and cracking of modules and cables

2. Solax's solution

In order to prevent the arcing of the DC side of the inverter from causing fires and other hazards, SolaX engineers have developed the integrated AFCI function, which detects the arcing of the DC side and cuts the circuit in time to protect the user and the electrical system. This function can reduce the probability of fire in the electrical system and reduce the damage of arc to the electrical system. So as to protect the personal and property safety of users.

The AFCI function will detect all series arcs within the DC side circuit from 200 to 750 J. When an arc is detected, the inverter stops running immediately and an error message is displayed within 2.5 seconds indicating that an arc fault has been detected. When the arcing disappears or the inverter detects that the arcing fault is a false alarm, the inverter will resume operation within 5 minutes automatically and start the detection of arcing again.

	SolaX Solution
Arc fault detection accuracy	99%
Max distance of arc detection	100m
Arc fault protection cycle-time	<2.5s
Arc energy range	200~750J

Automatic reset time	<5min
Max current of arc detection	16 A

The following inverters are available with the optional integrated AFCI function:

Inverter Type	Model	Availability
On-grid inverter	X1-MINI G4	Optional
	X1-BOOST G4	Optional
	X3-MIC G2	Optional
	X3-PRO G2	Optional
	X3-MEGA G2	Optional
	X3-FORTH	Optional
	X3-FORTH PLUS	Optional
	X3-GRAND	Optional
Energy storage inverter	X3-IES	Optional
	X1-Hybrid-G4	Developing
	X3-Hybrid-G4	Optional
	X3-HYB-G4 PRO	Optional
	X1-Vast 5-10kW	Optional
	X3-ULTRA	Optional
	J3-ULTRA-LV	Optional
	J3-ULTRA	Optional
	A1-Hybrid-G2	Yes